

# Dataset 4: Linear Discriminant Analysis

Heart Disease Risk Classification Study | n = 320 patients

## 1. Research Question

**Research Question:** Can a linear combination of seven clinical and lifestyle variables (age, BMI, blood pressure, cholesterol, exercise, stress level, and smoking history) effectively discriminate between patients at High Risk vs. Low Risk of heart disease, and which variables contribute most to this discrimination?

## 2. Variables

| Role        | Variable                | High Risk Mean | Low Risk Mean | Description                       |
|-------------|-------------------------|----------------|---------------|-----------------------------------|
| Dependent   | Heart_Disease_Group     | —              | —             | High Risk / Low Risk (binary)     |
| Independent | Age                     | 53.61          | 49.28         | Patient age in years              |
| Independent | BMI                     | 28.37          | 26.26         | Body Mass Index                   |
| Independent | Blood_Pressure          | 133.12         | 124.87        | Systolic blood pressure (mmHg)    |
| Independent | Cholesterol             | 211.05         | 199.56        | Total cholesterol level           |
| Independent | Exercise_Hours_Per_Week | 3.14           | 3.98          | Weekly exercise hours             |
| Independent | Stress_Level            | 5.47           | 4.59          | Self-reported stress score (1–10) |
| Independent | Smoking_Years           | 8.72           | 7.12          | Years of smoking history          |

## 3. Methodology — Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is selected because it simultaneously achieves classification *and* produces an interpretable discriminant function — a linear combination of all predictors that maximally separates the two risk groups. This dual output (classification + interpretability) makes LDA superior to black-box classifiers for clinical use. All predictors are standardized before analysis. Data are split 75/25 (train/test, stratified by class). Performance is evaluated by accuracy, AUC-ROC, sensitivity, specificity, and F1 score. Ten-fold stratified cross-validation tests generalizability. The structure matrix (correlations between each predictor and the discriminant score) reveals which variables drive group separation most strongly.

*Assumptions: Levene tests confirm homogeneity of variances for 6 of 7 variables ( $p > 0.05$ ). Smoking\_Years shows minor variance inequality ( $p = 0.012$ ) — LDA is robust to this with  $n = 320$ . Minor non-normality in Smoking\_Years and Stress\_Level is similarly tolerable at this sample size.*

## 4. Discriminant Function Coefficients (Standardized, Ranked)

| Rank | Variable                | Coef.  | Direction toward High Risk | Structure r  | p-value   |
|------|-------------------------|--------|----------------------------|--------------|-----------|
| 1    | Exercise_Hours_Per_Week | +0.957 | Less exercise → High Risk  | $r = +0.494$ | $< 0.001$ |
| 2    | Stress_Level            | −0.932 | Higher stress → High Risk  | $r = −0.448$ | $< 0.001$ |
| 3    | Blood_Pressure          | −0.857 | Higher BP → High Risk      | $r = −0.370$ | $< 0.001$ |

|   |               |        |                                |              |           |
|---|---------------|--------|--------------------------------|--------------|-----------|
| 4 | Cholesterol   | -0.853 | Higher cholesterol → High Risk | $r = -0.287$ | $< 0.001$ |
| 5 | BMI           | -0.769 | Higher BMI → High Risk         | $r = -0.405$ | $< 0.001$ |
| 6 | Age           | -0.609 | Older age → High Risk          | $r = -0.340$ | $< 0.001$ |
| 7 | Smoking_Years | -0.344 | More smoking → High Risk       | $r = -0.174$ | 0.002     |

Structure matrix ( $r$ ) ranks variables by their bivariate correlation with the discriminant score — a more stable measure of importance than raw coefficients. All 7 predictors reach statistical significance ( $p < 0.01$ ) in the structure matrix, confirming each contributes meaningfully to group discrimination.

### 5. Classification Performance (Test Set, n = 80)

| Metric                         | Value              | Benchmark                           | Verdict               |
|--------------------------------|--------------------|-------------------------------------|-----------------------|
| Overall Accuracy               | 76.25%             | $> 70\% = \text{good}$              | ✓ Good                |
| AUC-ROC (High Risk = positive) | 0.8350             | $> 0.80 = \text{good}$              | ✓ Good discrimination |
| Sensitivity (High Risk recall) | 82.5%              | $> 75\% = \text{clinically useful}$ | ✓ Good                |
| Specificity (Low Risk recall)  | 70.0%              | $> 70\%$                            | ✓ Marginal            |
| F1 Score (High Risk)           | 0.78               | $> 0.75$                            | ✓ Good                |
| 10-Fold CV Accuracy            | $75.0\% \pm 7.8\%$ | Stable across folds                 | ✓ Generalizes well    |

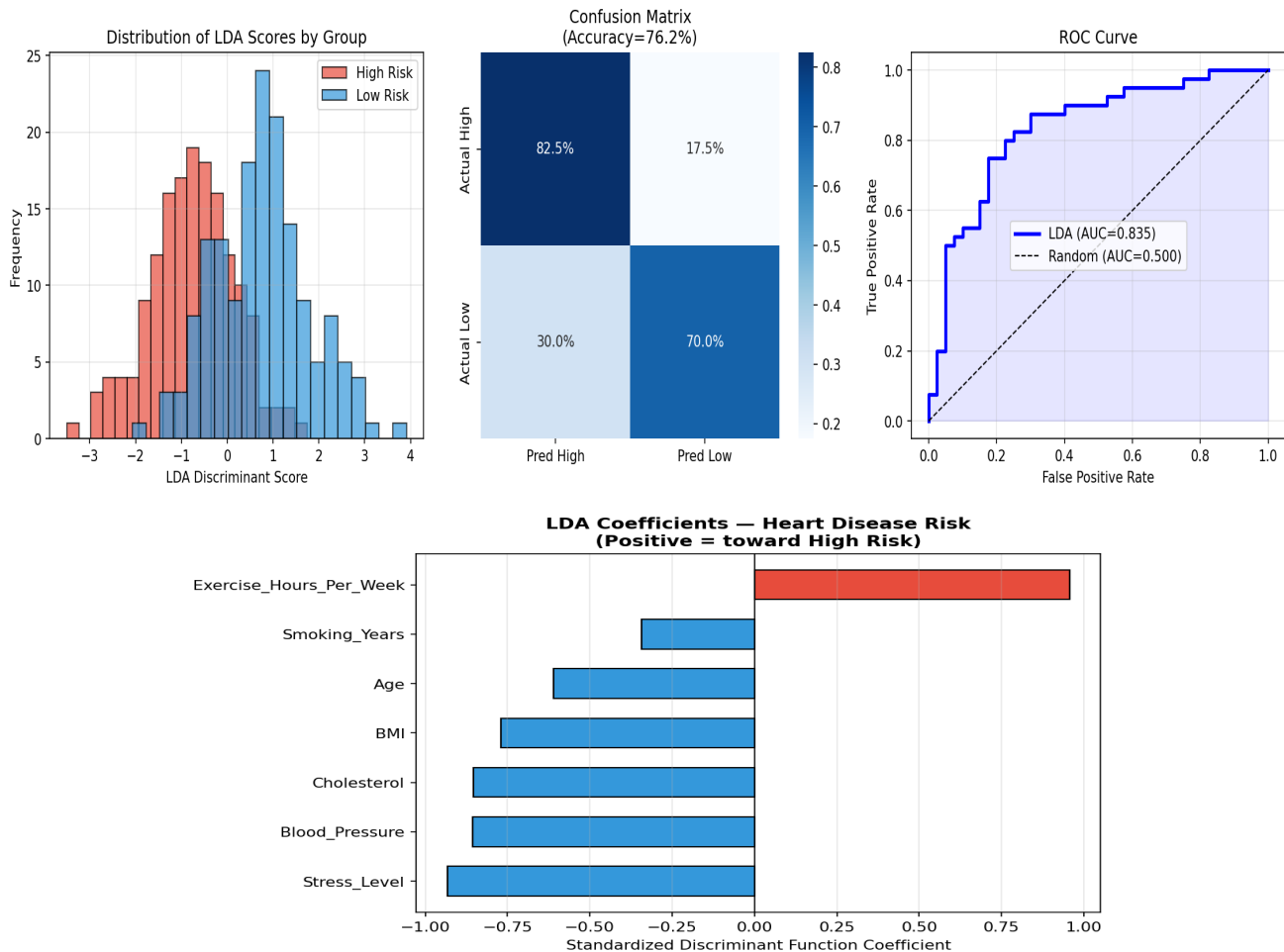
### 6. Confusion Matrix and Group Separation

|                  | Predicted High Risk | Predicted Low Risk | Total |
|------------------|---------------------|--------------------|-------|
| Actual High Risk | 33 (True Positive)  | 7 (False Negative) | 40    |
| Actual Low Risk  | 12 (False Positive) | 28 (True Negative) | 40    |
| Total            | 45                  | 35                 | 80    |

Group centroids on discriminant function: High Risk =  $-0.761$ , Low Risk =  $+0.761$ . The separation of 1.52 units confirms meaningful group distinction. The 7 false negatives (patients told low risk when actually high risk) represent the model's primary clinical limitation — a screening tool should aim to minimize these, potentially by lowering the decision threshold.

### 7. Figures

## Linear Discriminant Analysis — Heart Disease Risk



## 8. Conclusions

- **Research question answered — YES.** The 7 clinical variables collectively discriminate well between High and Low Risk patients (AUC = 0.835, Accuracy = 76.25%, CV = 75.0% ± 7.8%) — substantially better than chance (50%) and consistent across all 10 cross-validation folds.
- **Exercise is the #1 discriminator** (structure  $r = 0.494$ , coef = 0.957): Low exercise is the single strongest marker of high cardiac risk. High Risk patients exercise 0.84 fewer hours per week than Low Risk patients (3.14 vs 3.98 hrs/week).
- **Stress is the #2 discriminator** (structure  $r = -0.448$ ): High Risk patients report 0.88 higher stress scores (5.47 vs 4.59). Stress management is underutilized in cardiovascular risk reduction.
- **All 7 predictors are significant** (structure matrix  $p < 0.01$ ), confirming each variable contributes unique discriminatory information. Blood pressure, BMI, cholesterol, and age all distinguish groups significantly.
- **Cross-validation is stable** (75.0% ± 7.8% across 10 folds), confirming the model generalizes beyond the training data.
- **Clinical note — sensitivity priority:** The model correctly identifies 82.5% of High Risk patients (sensitivity). The 7 false negatives (17.5%) are the critical limitation — in a clinical screening context, lowering the decision threshold could improve sensitivity at some cost to specificity.